
Data With the Stars: Inferring Hidden Fan Support Under Voting-Rule Constraints

Dancing with the Stars combines expert evaluation with audience voting, but the fan-vote component is secret. This makes Problem C fundamentally different from a direct prediction problem: the observed eliminations do not reveal a unique set of fan votes. Our solution therefore treats fan support as a latent variable. Instead of pretending to know exact vote totals, we reconstruct weekly competitions, translate observed eliminations into feasible fan-support constraints, estimate maximum-entropy fan-share distributions, and use those estimates to compare voting rules and propose a producer-facing rule.

Using the official data set of 421 contestants across 34 seasons, we rebuilt 2,777 active contestant-week records and modeled 261 elimination weeks. The resulting latent-support model exactly reproduced the observed modeled eliminations by construction and confirmed that the underlying fan-vote uncertainty is large. This non-identifiability is the central practical finding: many fan-support profiles can explain the same observed elimination. It means that a good voting system should not merely chase an estimated “true” fan vote, but should be robust to close and ambiguous cases.

Our counterfactual simulation shows that rank and percent rules disagree in 26.4% of modeled elimination weeks, while estimated fan support changes the judge-only bottom set in 63.6% of modeled weeks. We also examine controversial contestants, professional-dancer and celebrity-category effects, and the institutional role of bottom-two judge saves. These results support a hybrid recommendation: use a percent-based combined score for transparency, but introduce a gray-zone judge save when the bottom-two combined margin is small. In our simulation, the gray-zone trigger activates in 61.7% of modeled elimination weeks, making close calls explicit rather than hiding them inside opaque vote totals.

For producers, the recommendation is simple: keep fan influence visible, avoid overclaiming exact fan support, and reserve judge discretion for statistically fragile bottom-two cases. The proposed system is more explainable than a pure rank system and more stable than treating a tiny combined-score gap as decisive.

1 Introduction

Dancing with the Stars (DWTS) is a useful voting-system case study because it combines two sources of evaluation that are deliberately different: judges score dance quality, while the public votes for contestants. The official Problem C data set reports contestant information, final outcomes, placements, and weekly judges' scores, but it does not reveal the fan votes. The hidden vote component creates the central modeling difficulty: a contestant can be eliminated under many different fan-vote profiles, and the same public outcome does not identify a unique distribution of fan support.

This observation changes the problem from a direct prediction task into an inverse problem. A supervised model that “predicts” fan votes would be unverifiable because no fan-vote labels exist in the data. Instead, we use the known voting systems as constraints. For each week, the observed elimination is treated as evidence that the true fan support must have belonged to a feasible set. We then estimate a conservative point inside that set and use the uncertainty itself as part of the answer.

Our approach has four linked layers. First, we reconstruct each weekly competition from the wide official data. Second, we formalize the rank, percent, and bottom-two judge-save mechanisms. Third, we estimate latent fan support using feasible-set constraints and a maximum-entropy principle. Fourth, we use the estimated support profiles for counterfactual simulations and a producer-facing voting recommendation. Figure 1 summarizes the complete workflow.

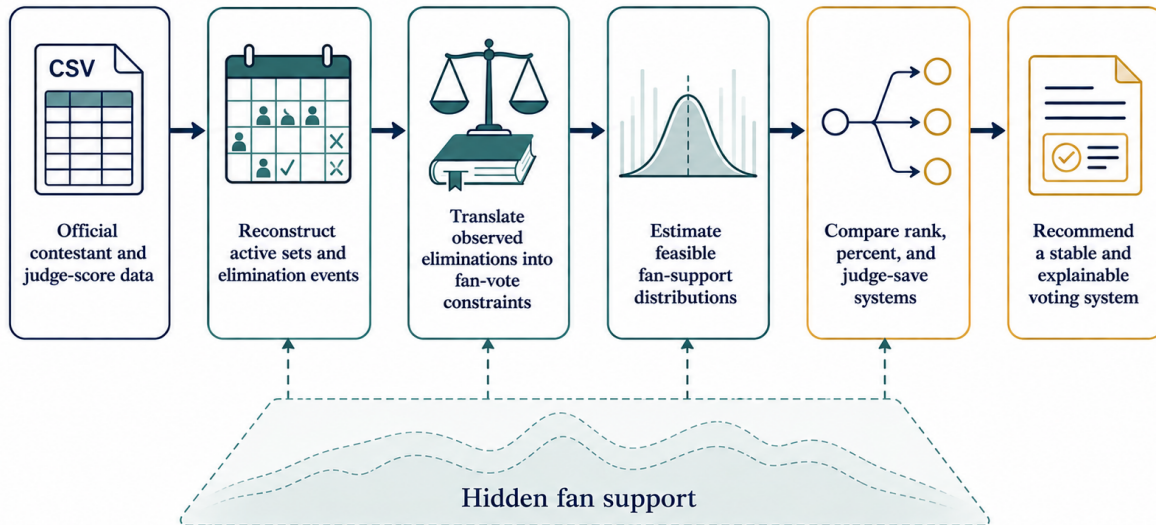


Figure 1: Overall research workflow. Fan support is treated as a hidden variable constrained by observed eliminations rather than as an observed training label.

The main contribution of the paper is not a claim that exact fan votes can be recovered. Rather, we show how to quantify what can be inferred from the publicly observable record, how different voting rules behave under the same latent support estimates, and how producers can design a rule that remains

transparent when the bottom of the leaderboard is statistically fragile.

1.1 Problem restatement

The official problem asks for a data-driven analysis of the DWTS voting system. We restate the modeling tasks as four operational questions.

1. Reconstruct each season-week competition from the official contestant data and identify the active set, judge scores, and observed eliminations.
2. Infer what can be learned about hidden fan support without assuming unobserved vote totals.
3. Compare plausible voting rules under the same judge scores and latent fan-support estimates, especially rank aggregation, percent aggregation, and bottom-two judge saves.
4. Recommend a transparent producer-facing voting system that balances expert judgment, viewer influence, and robustness in close elimination weeks.

These questions deliberately separate historical reconstruction from policy recommendation. A model that exactly reproduces past eliminations is useful as a consistency check, but the design decision for producers depends on how rules behave when the bottom-two margin is small and hidden fan support is non-identifiable.

2 Notation and Assumptions

Table 1: Core notation used in the modeling workflow.

Symbol	Type	Meaning
s	index	DWTS season.
t	index	Competition week within a season.
i	index	Active contestant in a season-week.
$A_{s,t}$	set	Contestants active in season s , week t .
$E_{s,t}$	set	Contestants eliminated after week t .
J_i	scalar	Total judge score for contestant i .
P_i^J	scalar	Judge-score share, $J_i / \sum_k J_k$.
F_i	scalar	Hidden fan-share variable.
C_i^{pct}	scalar	Percent-rule combined score.
C_i^{rank}	scalar	Rank-rule combined score.
τ	scalar	Gray-zone threshold for bottom-two margins.

Assumption 1 (Official-data sufficiency). *The official CSV is treated as the authoritative source for contestant identity, weekly judge scores, placements, and final results. We do not add unverified external popularity data because fan votes are not observed in the official record.*

Assumption 2 (Active-set reconstruction). *A contestant is active in a week if at least one valid judge score is present and the weekly judge total is positive. Once a contestant disappears from the active set, the disappearance is interpreted as an elimination unless the whole season has ended.*

Assumption 3 (Rule-consistency inference). *Observed eliminations are assumed to be consistent with the reported judge scores and a hidden fan-support vector under a percent-rule feasible set. This assumption is not a claim that DWTS always used an identical rule; it is the common constraint system used to compare historical outcomes and candidate rules on a shared scale.*

Assumption 4 (No causal claim from observational effects). *Professional-dancer, industry, age, and survival models are interpreted as association models. They can identify recurring patterns in the data, but they cannot prove that a dancer, age, or celebrity category caused a specific elimination outcome.*

3 Data Audit and Preprocessing

3.1 Raw data audit

The official CSV contains 421 contestant records, 53 columns, and 34 seasons. For each contestant, the data include identity variables, celebrity industry, home location, age during season, final result, placement, and four possible judge-score columns for each week from Week 1 to Week 11. The raw file is a wide table: one row is one contestant, while weekly scores appear in repeated columns. This layout is convenient for storage but not for modeling a weekly elimination system.

We therefore first audited the data before fitting any model. Missing score cells can have several meanings: a fourth judge may not exist in a given week, a season may not include that week, or a contestant may already have been eliminated. In contrast, explicit zeros after a contestant leaves the competition should not be interpreted as poor performance by an active contestant. This distinction is important because the fan-vote constraints are constructed week by week from the active set.

Table 2: Core data audit metrics used by the modeling workflow.

Metric	Value
Contestant records	421
Raw columns	53
Seasons	34
Reconstructed panel rows	4,631
Active contestant-week rows	2,777
Reconstructed elimination-event rows	343
Single-elimination rows	221
Multi-elimination rows	82
No-elimination rows	40

3.2 Panel reconstruction

We converted the wide table into a long panel indexed by season, week, and contestant. For each contestant-week we computed the available judge count, judge total, judge mean, and an active-status flag. A contestant is active in a week if at least one judge score is present and the weekly judge total is positive. This rule avoids treating post-elimination zeros as active competition records.

Next, we reconstructed the weekly active sets and observed elimination events. Let $A_{s,t}$ be the active set in season s and week t . A contestant in $A_{s,t} \setminus A_{s,t+1}$ is recorded as eliminated after week t . If this set contains multiple contestants, the week is marked as a multi-elimination event. If the active set does not shrink, the week is marked as a no-elimination week. This reconstruction produced 2,777 active contestant-week rows and 261 modeled elimination weeks used by the fan-support inference model.

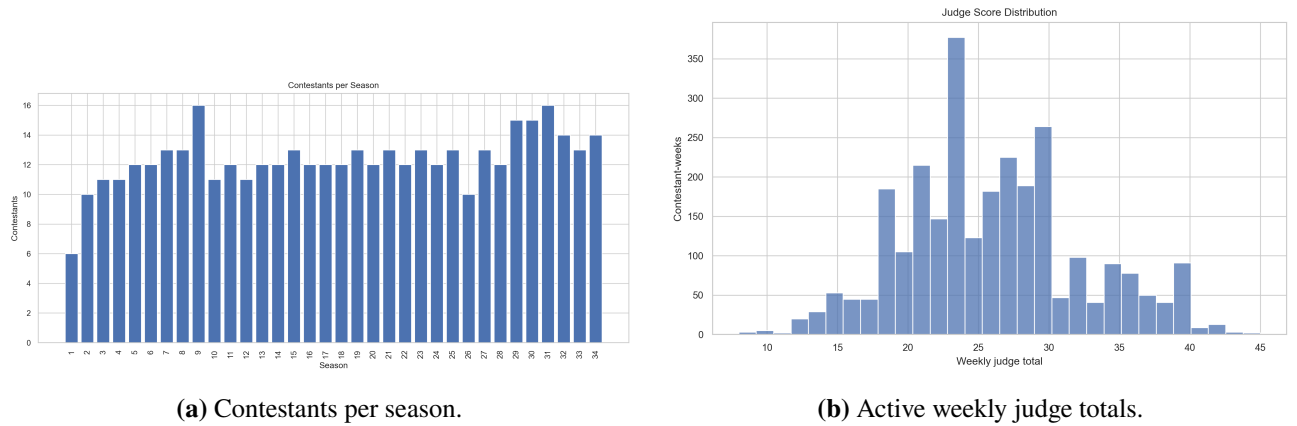


Figure 2: Data reconstruction diagnostics. Season sizes and judge-score distributions vary enough that all modeling must be performed on reconstructed season-week active sets.

3.3 Reconstruction diagnostics

The reconstructed active-set matrix is a useful diagnostic because many later calculations depend on having the right contestants in the right week. A season with a missing or duplicated active block would produce visible irregularities in Figure 3. The heatmap shows the expected staircase pattern: seasons begin with larger casts and contract over time, while a small number of weeks remain flat because no elimination occurs.

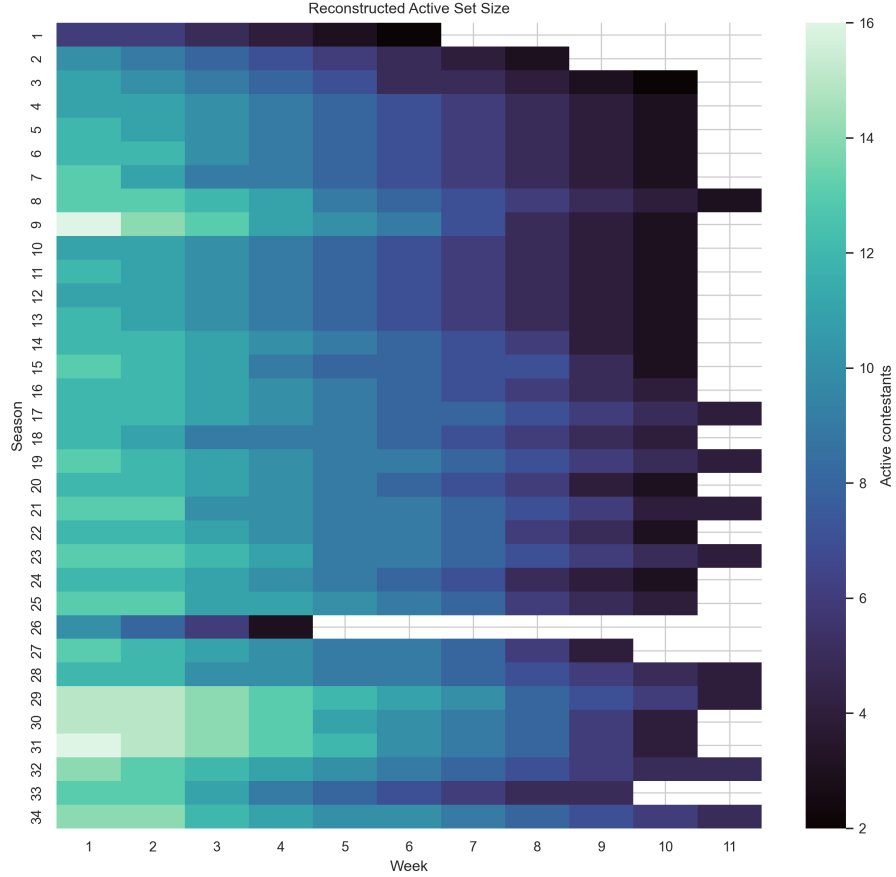


Figure 3: Reconstructed active contestants by season and week. The heatmap verifies that the panel transformation preserves the shrinking competition structure needed for week-level voting analysis.

This diagnostic matters because fan-support inference is local to a season-week. A contestant excluded from $A_{s,t}$ would not receive a fan-share variable, while a contestant incorrectly kept active after elimination would distort the simplex constraint $\sum_i F_i = 1$. We therefore treat active-set reconstruction as a model input rather than a bookkeeping detail.

4 Voting Rule Formalization

The contest history involves several ways of combining judges and viewers. We formalize three mechanisms: a rank rule, a percent rule, and a bottom-two judge-save rule. Figure 4 gives the conceptual comparison.

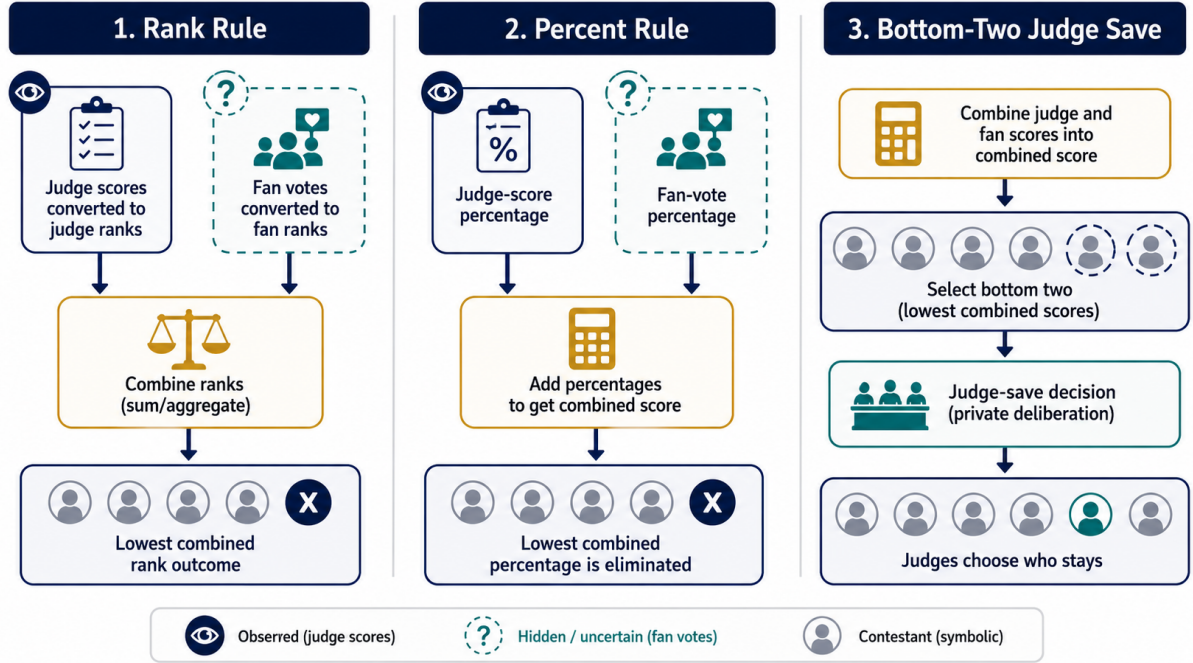


Figure 4: Mechanism comparison for rank, percent, and bottom-two judge-save systems. The fan component is hidden and must be inferred from constraints.

For season s , week t , and active contestant $i \in A_{s,t}$, let J_i be the observed total judge score and let F_i be the unknown fan-share variable. We normalize fan shares so that

$$F_i \geq 0, \quad \sum_{i \in A_{s,t}} F_i = 1.$$

Under a percent rule, judge scores and fan shares are both expressed as shares. The normalized judge share is

$$P_i^J = \frac{J_i}{\sum_{k \in A_{s,t}} J_k},$$

and the combined score is

$$C_i^{\text{pct}} = P_i^J + F_i.$$

The lowest combined score is eliminated. If the observed eliminated set for a week is $E_{s,t}$ and the survivors are $S_{s,t} = A_{s,t} \setminus E_{s,t}$, then every eliminated contestant must have combined score no larger than every survivor:

$$P_e^J + F_e \leq P_j^J + F_j, \quad e \in E_{s,t}, \quad j \in S_{s,t}.$$

These inequalities define a feasible fan-support polytope.

Under a rank rule, the judge score and fan support are converted into ranks:

$$C_i^{\text{rank}} = R_i^J + R_i^F,$$

where larger C_i^{rank} means a worse combined rank. The contestant or contestants with the worst combined rank are eliminated. We use the maximum-entropy fan-share estimate to induce fan ranks for rank-rule counterfactuals.

The bottom-two judge-save mechanism is represented as a two-stage process. First, a combined score identifies the bottom two contestants. Second, judges choose which contestant to save. Because detailed judge-save votes are not fully observed in the data, we use it as a policy simulation rather than a direct historical reconstruction.

Table 3: Voting-rule formalization used in the simulations.

Rule	Score	Elimination decision
Percent	judge share + fan share	lowest combined percent
Rank	judge rank + fan rank	worst combined rank
Bottom-two judge save	combined score, then judge decision	judge save among bottom two

5 Fan Vote Estimation

5.1 Why fan votes are not directly predictable

The fan vote is hidden and non-identifiable from the observed data alone. If a contestant is eliminated, we learn that the combined judge-fan result placed that contestant at or near the bottom under the relevant rule. We do not learn the exact vote total, the absolute number of votes, or the unique share of each contestant. This is why our model estimates feasible support rather than exact votes.

5.2 Feasible fan-support intervals

For each modeled elimination week, we build the percent-rule linear constraints from the observed eliminated set. We then solve two linear programs for each contestant i :

$$\min F_i \quad \text{and} \quad \max F_i$$

subject to the simplex constraints and the elimination inequalities above. The result is an interval $[F_i, \bar{F}_i]$ that describes how much fan support contestant i could have had while still being consistent with the observed elimination. We used SciPy’s linear programming solver for this step [6].

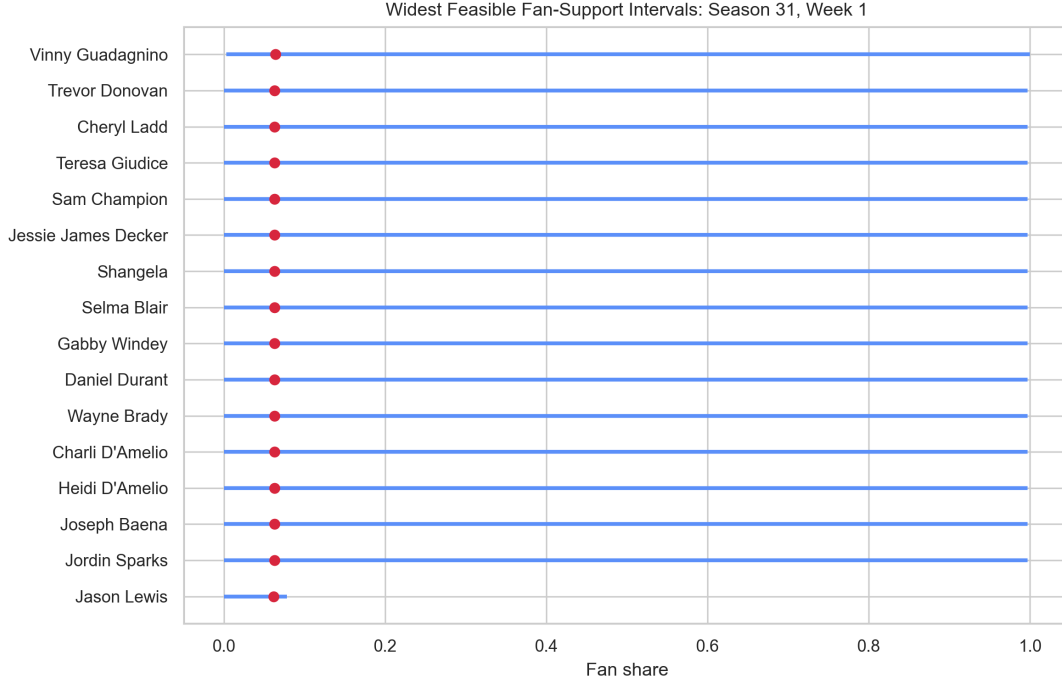


Figure 5: Example of feasible fan-support intervals for a high-uncertainty season-week. The red points show maximum-entropy estimates inside wide feasible intervals.

5.3 Maximum-entropy point estimate

Intervals are useful for uncertainty, but counterfactual simulations also need a single support profile. We choose the maximum-entropy feasible profile:

$$\max_F - \sum_{i \in A_{s,t}} F_i \log F_i$$

subject to the feasible-set constraints. The entropy objective selects the least concentrated fan-support vector consistent with the observed outcome. It therefore avoids arbitrarily inventing a dramatic fan favorite unless the elimination constraints require one. This follows the maximum-entropy principle introduced by Shannon [3] and is implemented as a convex optimization problem using CVXPY [5].

Table 4: Fan-support inference algorithm used for each modeled elimination week.

Step	Operation
1	Build the active set $A_{s,t}$ from reconstructed contestant-week records.
2	Normalize judge totals into judge shares P_i^J .
3	Translate each observed eliminated-survivor pair into a linear inequality $P_e^J + F_e \leq P_j^J + F_j$.
4	Solve linear programs for each contestant's feasible lower and upper fan-share bounds.
5	Solve the maximum-entropy convex program inside the feasible polytope.
6	Use the resulting support vector for validation, rule counterfactuals, controversial-case diagnostics, and the proposed-system simulation.

The algorithm produces three different outputs from the same constraints: intervals, a point estimate, and diagnostic flags. This is important because a single estimated fan-share vector should not be treated as ground truth. The intervals tell us how much freedom remains after the historical elimination is enforced, while the maximum-entropy vector gives a stable reference point for comparing rules.

5.4 Uncertainty as a model output

The average feasible interval width across modeled weeks is 0.862, which is large on the $[0, 1]$ fan-share scale. This does not mean the model is weak; it means the historical outcomes do not uniquely identify fan votes. That uncertainty is itself a substantive result. It warns against overinterpreting any one estimated fan-support profile and motivates a voting rule that handles close bottom-two cases explicitly.

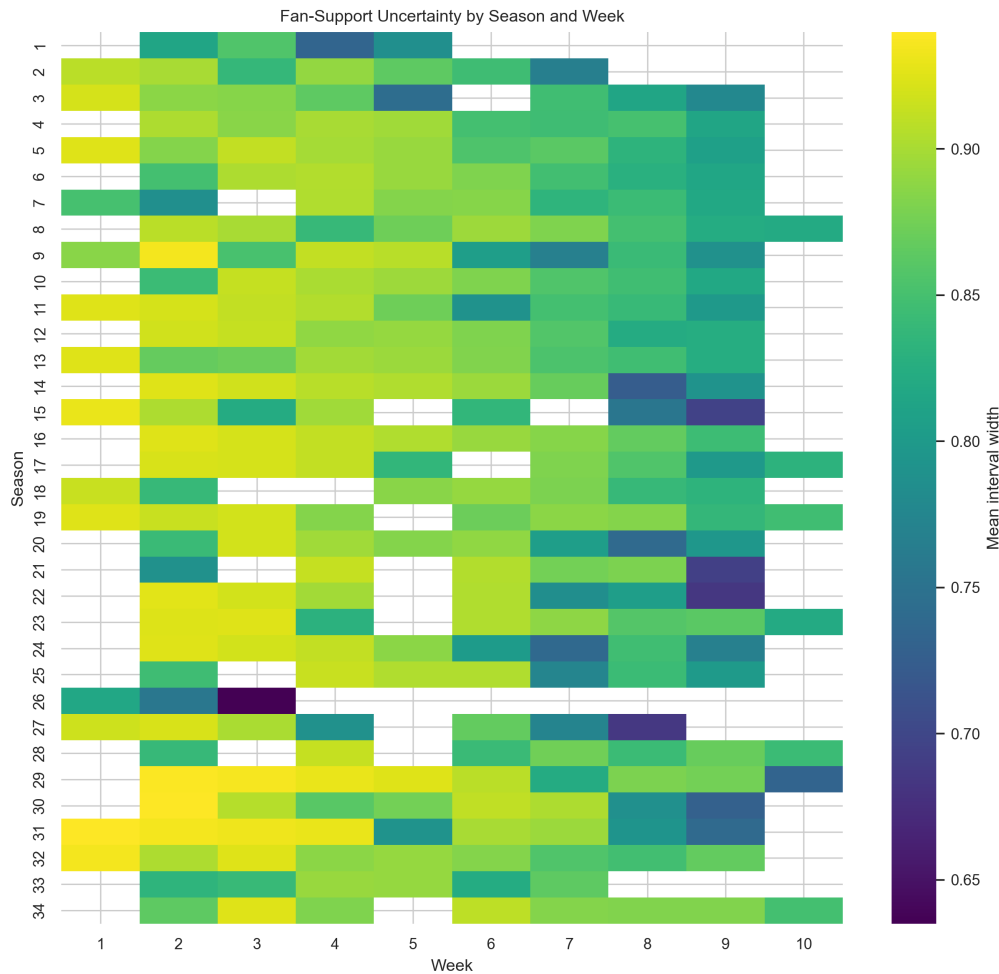


Figure 6: Fan-support uncertainty by season and week. Wide feasible intervals indicate that many latent fan-vote profiles can produce the same observed elimination.

6 Model Validation and Baseline Tests

The validation step asks a limited but essential question: after reconstructing the active set and solving for a feasible maximum-entropy support vector, does the model reproduce the observed elimination under the percent-rule constraints? The answer is yes for all 261 modeled elimination weeks. This 100% reproduction rate should not be misread as out-of-sample prediction. It is a consistency check showing that the reconstructed data, constraints, and optimization pipeline agree with the observed historical record.

Table 5: Validation and uncertainty metrics.

Metric	Value
Modeled elimination weeks	261
Percent-rule reproduction rate	100.0%
Observed elimination in modeled bottom two	100.0%
Mean feasible fan-share interval width	0.862

The more important validation result is the scale of uncertainty. The average interval width of 0.862 shows that observed eliminations alone leave the fan support profile highly underdetermined. In practical terms, a reported elimination may be consistent with a wide range of public-vote configurations. This strengthens the case for comparing voting systems through robust counterfactual behavior rather than through a single supposed true vote count.

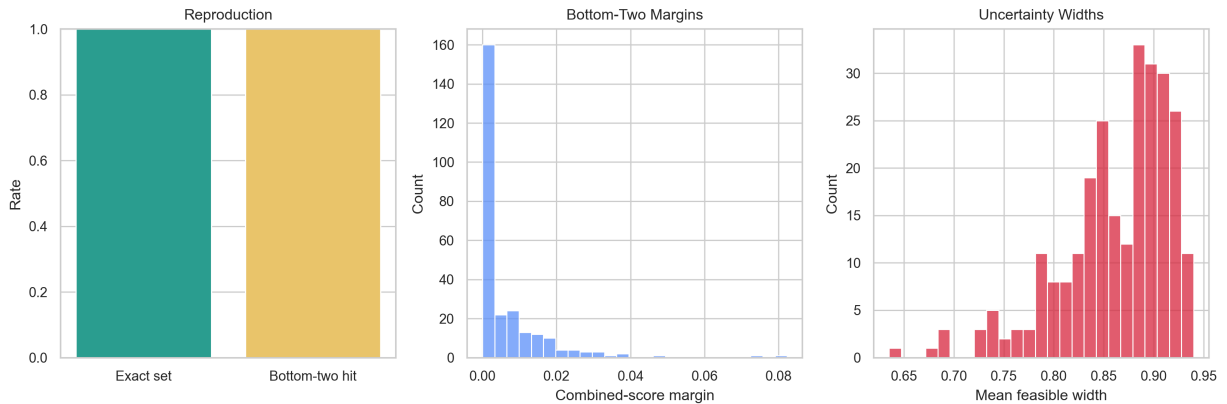


Figure 7: Validation dashboard. The model reproduces the observed modeled eliminations, while the margin and interval-width panels show that close and uncertain weeks are common.

6.1 Baseline comparison

The consistency check above is necessary but not sufficient. Because the percent-rule fan vector is constructed to satisfy the observed elimination constraints, exact reproduction should be interpreted as feasibility rather than predictive accuracy. We therefore compare it with three simpler baselines: a rank-rule counterfactual using the same inferred fan support, a judge-only bottom set, and a fan-only bottom set.

Table 6: Baseline and counterfactual rule comparison over 261 modeled weeks.

Candidate rule	Exact match	Mean overlap	Interpretation
Percent + inferred fan	1.000	1.000	constraint reconstruction
Rank + inferred fan	0.736	0.760	rule-change counterfactual
Fan-only bottom	0.625	0.681	audience-only baseline
Judge-only bottom	0.364	0.416	no-fan baseline

The judge-only baseline matches the historical eliminated set in only 36.4% of modeled weeks, while the fan-only bottom set matches 62.5%. This does not mean that fan preference alone determined eliminations; judge scores still shape the feasible set and the combined score. It does show that hidden fan support is not a small perturbation around the judge leaderboard. The rank-rule counterfactual matches 73.6% of historical eliminated sets, leaving more than one quarter of weeks where the aggregation rule itself matters.

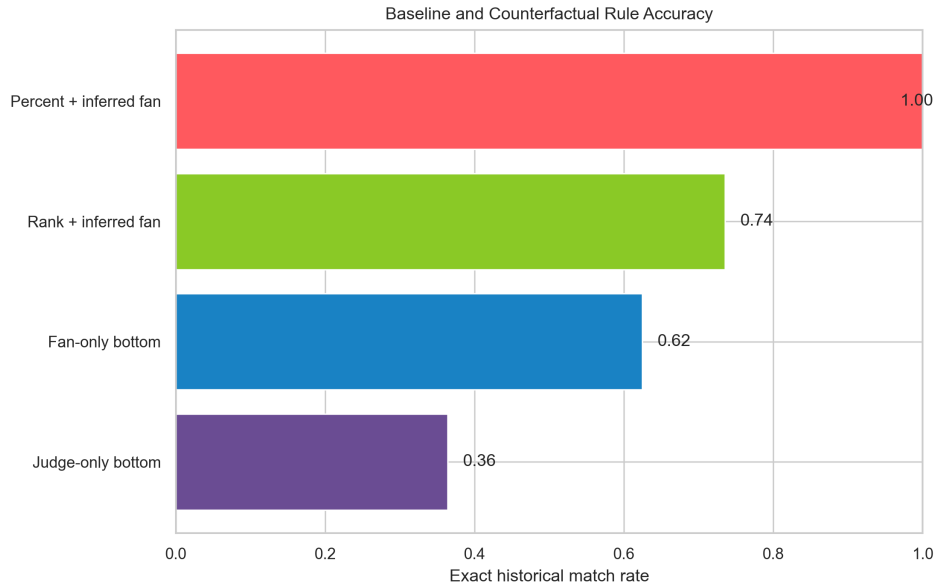


Figure 8: Baseline and counterfactual exact-match rates. The percent model is a reconstruction benchmark, while the other bars show how much is lost when the same data are forced through simpler voting assumptions.

7 Rank vs. Percent Counterfactual Analysis

We next use the same maximum-entropy fan-support estimates to compare the rank and percent rules. This isolates the effect of the rule itself: the contestants, judge scores, and estimated fan support remain fixed, while only the rule used to combine them changes.

Across 261 modeled elimination weeks, rank and percent rules produce different elimination sets in 26.4% of weeks. Estimated fan support changes the judge-only bottom set in 63.6% of weeks. These two metrics have different interpretations. The first measures institutional sensitivity to rule choice.

The second measures how often the public component is strong enough to change what would have happened under judges alone.

Table 7: Counterfactual rule-comparison metrics.

Metric	Value
Modeled elimination weeks	261
Rank-percent outcome difference rate	26.4%
Fan override rate relative to judge-only bottom set	63.6%
Mean rank-percent eliminated-set overlap	0.893

The heatmap in Figure 9 shows that rule disagreement is not confined to a single season. It appears throughout the history of the data, which means that voting-rule design is not a minor technicality. A rule change can alter which contestants are at risk even when the same hidden support estimate is used.

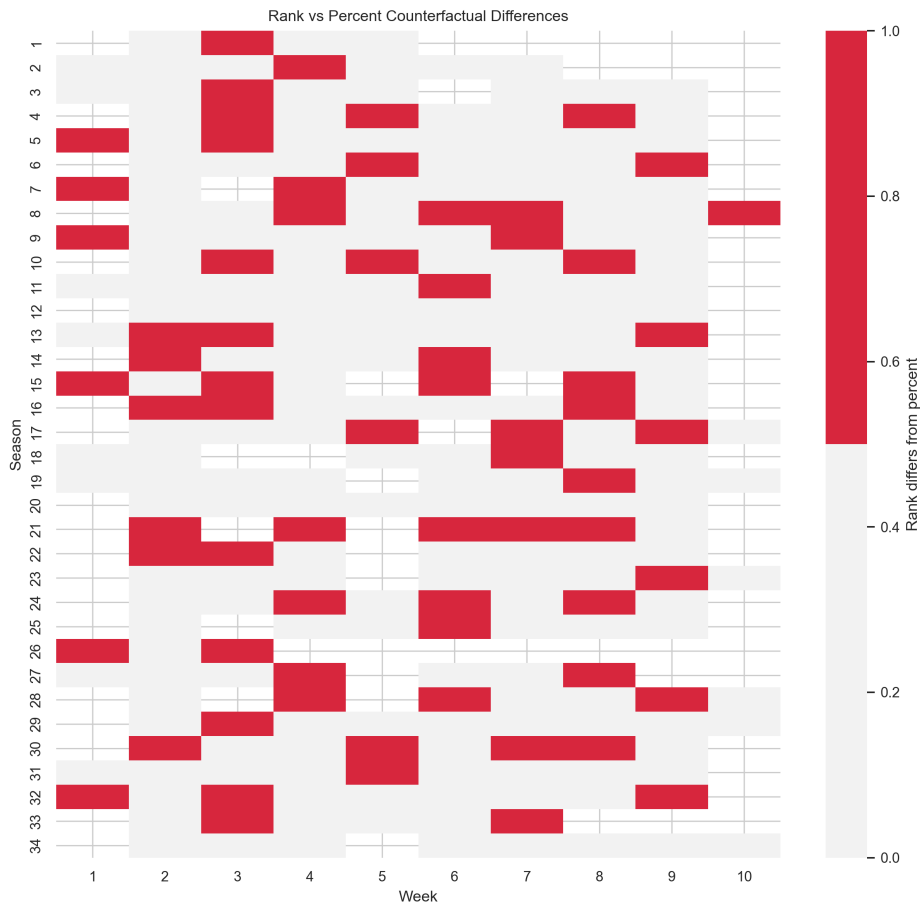


Figure 9: Season-week map of rank-versus-percent differences. Dark cells mark modeled weeks where the two rules select different eliminated sets.

7.1 High-leverage rule-switch weeks

The most informative counterfactuals are not the average weeks but the weeks where the rule changes the at-risk contestants. Table 8 lists representative cases with large uncertainty and no overlap between the rank-rule and percent-rule eliminated sets. Season 9, Week 1 is especially diagnostic because it is a multi-elimination week: the percent rule reproduces Ashley Hamilton and Macy Gray as the eliminated pair, while the rank rule would instead place Chuck Liddell and Debi Mazar at the bottom.

Table 8: Representative rank-versus-percent switch cases.

Season	Week	Observed	Percent rule	Rank rule	Mean width
9	1	Ashley Hamilton; Macy Gray	Macy Gray; Ashley Hamilton	Chuck Liddell; Debi Mazar	0.885
30	2	Martin Kove	Martin Kove	Matt James	0.940
29	3	Carole Baskin	Carole Baskin	Justina Machado	0.936
32	1	Matt Walsh	Matt Walsh	Alyson Hannigan	0.935
15	1	Pamela Anderson	Pamela Anderson	Kirstie Alley	0.930

These cases explain why percent aggregation is preferable for transparency. In a rank system, small score differences can be converted into large rank jumps, especially early in a season when many contestants are active. Percent scoring keeps the magnitude of judge-score differences visible.

7.2 Fan-rescue diagnostics

We define a fan-rescue gap as

$$G_{i,t} = R_{i,t}^J - R_{i,t}^F.$$

A large positive value means that the contestant ranked poorly with judges but well with the inferred audience profile. The largest estimated rescue case is Vinny Guadagnino in Season 31, Week 1, whose judge rank was 16 while his maximum-entropy fan rank was 1.

Table 9: Largest estimated fan-rescue contestant-weeks.

Season	Week	Contestant	Judge rank	Fan rank	Gap	Result
31	1	Vinny Guadagnino	16	1	15	Eliminated Week 8
31	2	Cheryl Ladd	15	1	14	Eliminated Week 3
30	3	Cody Rigsby	14	1	13	3rd Place
31	3	Vinny Guadagnino	14	1	13	Eliminated Week 8
9	1	Michael Irvin	14	1	13	Eliminated Week 7

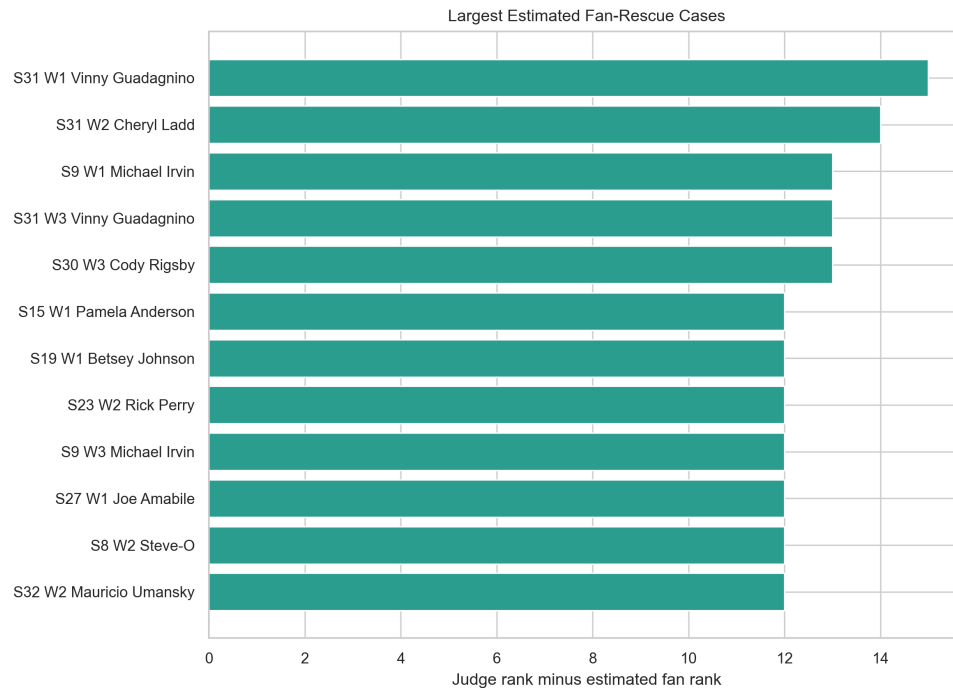


Figure 10: Largest estimated fan-rescue gaps. Positive gaps show weeks where inferred audience support substantially offset weak judge ranking.

8 Controversial Contestants

The problem statement highlights cases where public popularity and judge scores may appear to pull in different directions. We therefore tracked Jerry Rice, Billy Ray Cyrus, Bristol Palin, and Bobby Bones using a controversy index,

$$CI_{i,t} = |R_{i,t}^J - R_{i,t}^F|,$$

where R^J is judge rank and R^F is the estimated fan rank. Large values indicate a large gap between expert evaluation and inferred public support.

Table 10: Summary of selected controversial contestants.

Contestant	Season	Weeks	Mean judge rank	Mean fan rank	Result
Bristol Palin	15	4	10.50	4.25	Eliminated Week 4
Bobby Bones	27	7	7.29	4.86	1st Place
Bristol Palin	11	9	6.22	3.89	3rd Place
Billy Ray Cyrus	4	7	5.86	4.00	Eliminated Week 8
Jerry Rice	2	7	4.86	4.14	2nd Place

The pattern is clearest for Bristol Palin's Season 15 run and Bobby Bones's Season 27 win. In both cases, estimated fan ranks are substantially better than judge ranks over multiple weeks. This does not

prove a specific vote total, but it does quantify the direction of the conflict: these contestants survived longer than judge rank alone would suggest because the feasible fan-support estimates consistently place them higher with viewers.

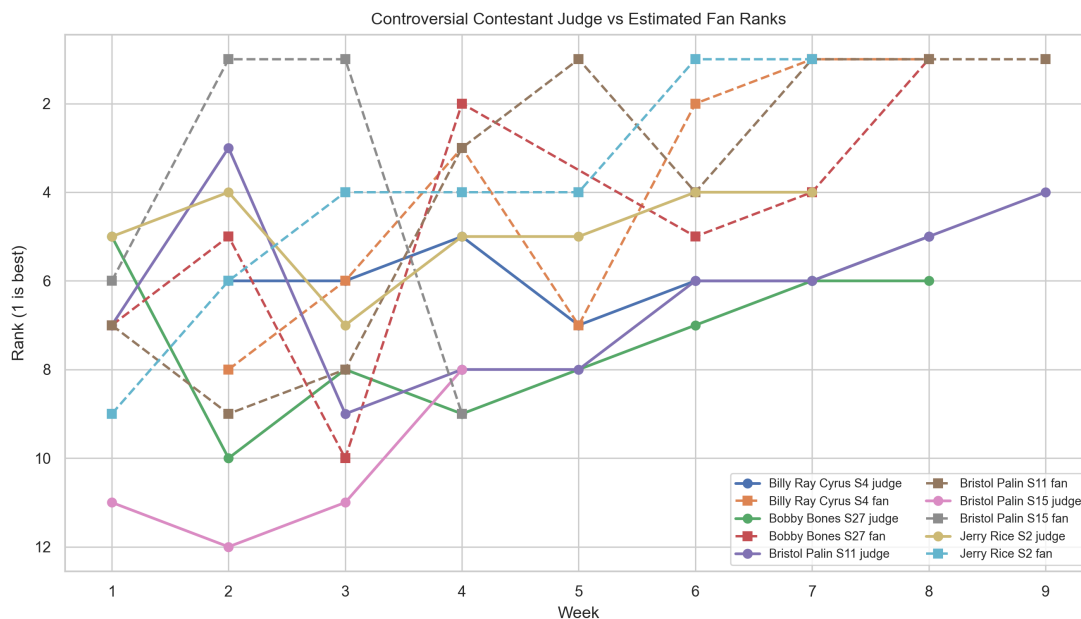


Figure 11: Judge rank versus estimated fan rank for selected controversial contestants. Lower rank values are better.

8.1 Data-driven controversy scan

The four highlighted names are useful because they are recognizable examples, but the model can also scan all contestants for judge-fan instability. Table 11 ranks contestants by the maximum weekly gap between judge rank and estimated fan rank, requiring at least three modeled weeks so that a single anomalous week does not dominate the list.

Table 11: Contestants with the largest judge-fan instability.

Contestant	Season	Weeks	Mean judge rank	Mean fan rank	Result
Vinny Guadagnino	31	8	10.88	4.50	Eliminated Week 8
Joanna Krupa	9	9	2.67	6.78	Eliminated Week 9
Cheryl Ladd	31	3	12.67	5.67	Eliminated Week 3
Michael Irvin	9	7	9.86	1.71	Eliminated Week 7
Cody Rigsby	30	8	8.12	4.25	3rd Place
Steve-O	8	5	10.20	2.00	Eliminated Week 6

This wider scan reveals two different kinds of controversy. Vinny Guadagnino, Michael Irvin, and Steve-O are fan-rescue cases: judges placed them relatively low, but the inferred audience ranks are

much stronger. Joanna Krupa is the opposite pattern: strong judge ranks combined with weaker inferred fan support. Both patterns are important to producers because viewer complaints can arise from either side of the judge-audience divide.

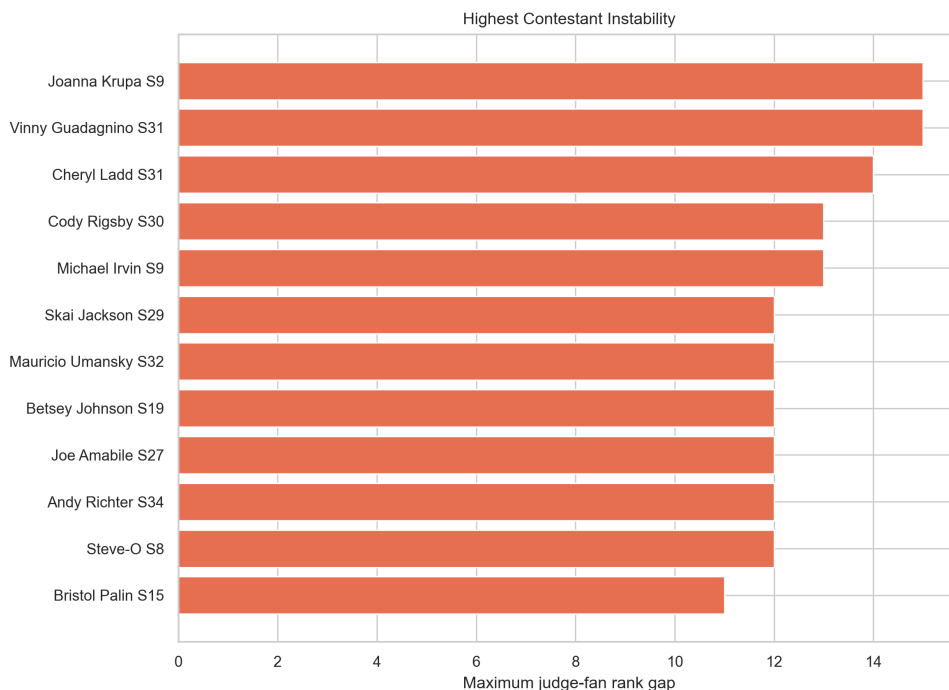


Figure 12: Contestants with the highest maximum judge-fan rank gap among those with at least three modeled weeks.

9 Professional Dancer and Celebrity Effects

We modeled three related effects: judge-score effects, fan-support effects, and survival-time effects. Judge and fan effects were estimated with ordinary least squares using celebrity age and industry indicators. Survival effects were estimated with a penalized Cox model using active-week duration and an elimination indicator. These models are exploratory rather than causal: the data are observational and contestants are not randomly assigned to industries or professional partners.

The strongest consistent result is that age is associated with weaker estimated fan support and shorter survival. In the fan-support model, celebrity age has a negative coefficient of -0.00051 per year. In the survival model, age has a positive hazard coefficient of 0.0327 , meaning older contestants tend to face earlier elimination after controlling for the included industry indicators. Some industry coefficients are also notable: for example, radio personalities and politicians receive lower judge-score estimates in the judge-score model, while social media personalities and TV personalities have lower estimated fan-share coefficients in the fan-support model. These findings should be interpreted as association patterns, not as causal claims.

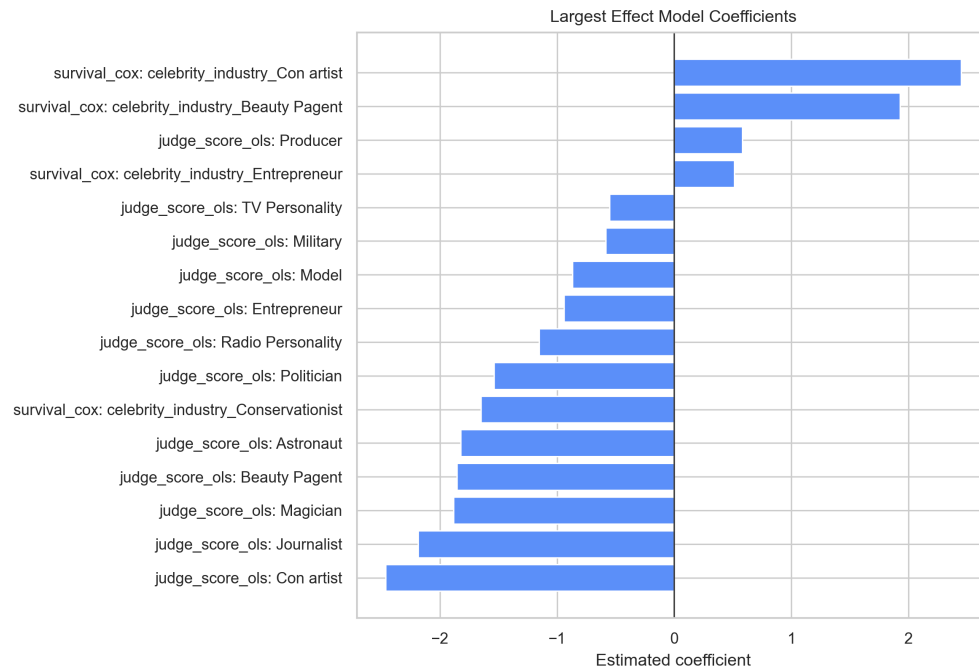


Figure 13: Largest coefficients from the exploratory judge-score, fan-support, and survival models. Coefficients summarize associations, not causal effects.

10 Proposed Voting System

The central design lesson is that close bottom-two cases should be treated as ambiguous rather than forced into a deterministic public-facing ranking. We therefore propose a hybrid system:

Percent Rule + Gray-Zone Judge Save.

The system first converts judge scores and fan votes into percentages and adds them. If the combined-score gap between the lowest two contestants is larger than a threshold τ , the lowest combined score is eliminated. If the margin is at or below τ , the bottom two enter a judge-save decision. In our simulation, τ is set to 0.00354, the 25th percentile of positive modeled bottom-two combined-score margins.

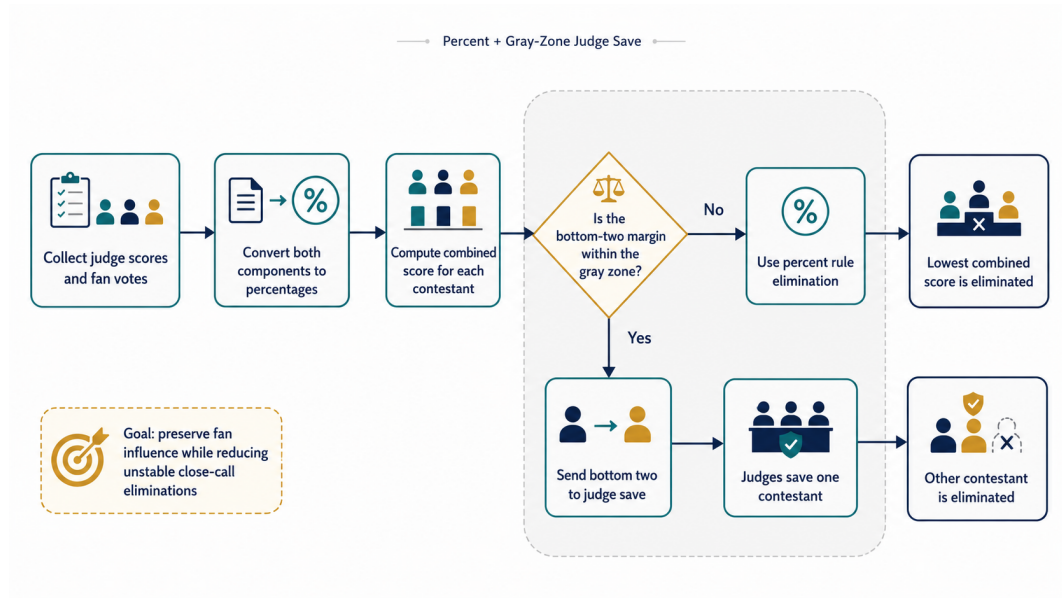


Figure 14: Proposed Percent + Gray-Zone Judge Save system. The judge save is activated only when the bottom-two combined-score margin is small.

The proposed system has three advantages. First, percent scoring is more transparent than rank aggregation because judge and fan shares are measured on the same scale. Second, the gray zone acknowledges that tiny margins are not substantively robust when fan support is hidden and highly non-identifiable. Third, judge discretion is limited to close bottom-two cases, which preserves fan influence while giving producers a defensible way to manage fragile eliminations.

Table 12: Simulation summary for the proposed gray-zone system.

Metric	Value
Gray-zone threshold τ	0.00354
Gray-zone trigger rate	61.7%
Proposed system differs from pure percent rule	46.0%
Proposed elimination appears in observed set	54.0%

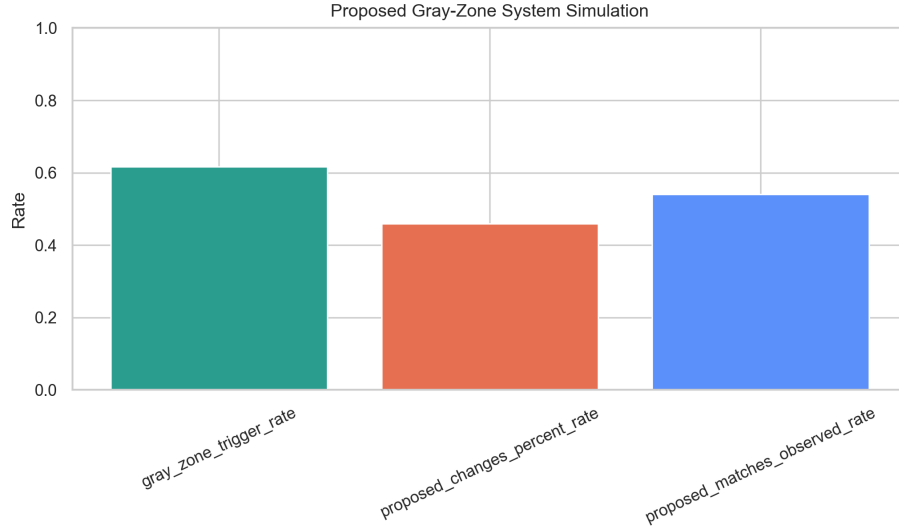


Figure 15: Simulation dashboard for the proposed gray-zone system.

11 Sensitivity Analysis

The proposed system depends on the gray-zone threshold τ , so we test how the recommendation changes when τ is chosen from different quantiles of the positive bottom-two margin distribution. The “No gray zone” row in Table 13 is the pure percent reconstruction. It matches the historical modeled eliminations by construction, but it offers no special treatment for statistically fragile close calls.

Table 13: Sensitivity of the proposed system to the gray-zone threshold.

Threshold	Trigger rate	Changes percent	Historical match
No gray zone	0.000	0.000	1.000
5th pct.	0.513	0.460	0.540
10th pct.	0.540	0.460	0.540
25th pct.	0.617	0.460	0.540
50th pct.	0.743	0.460	0.540
75th pct.	0.870	0.460	0.540
90th pct.	0.946	0.460	0.540

The sensitivity curve has a useful interpretation. Once the threshold is large enough to activate the judge save in the genuinely close cases, the set of weeks where the proposed system differs from pure percent stabilizes at 46.0%. Increasing τ further sends more weeks to the gray-zone procedure, but many of those additional weeks still eliminate the same contestant because the judge-weaker bottom-two contestant is already the percent-rule loser. We use the 25th percentile threshold as a middle setting: it captures close calls without turning the judge save into the default elimination mechanism.

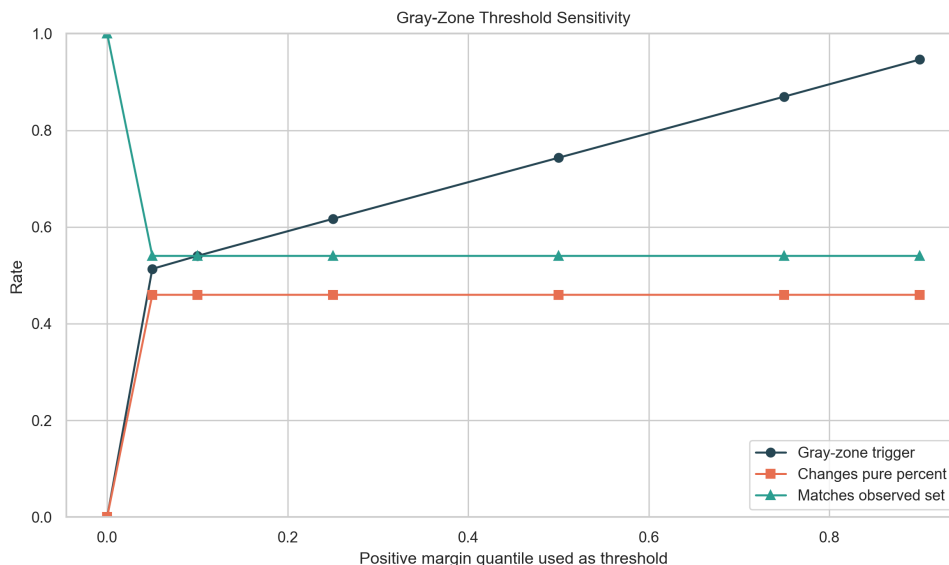


Figure 16: Gray-zone threshold sensitivity. The 25th percentile setting gives a moderate trigger rate while preserving a stable set of proposed changes relative to pure percent aggregation.

11.1 Robustness of the main conclusions

The major conclusions do not depend on a single threshold value. First, hidden fan support remains highly non-identifiable across all settings; the mean feasible fan-share interval width is 0.862 before the gray-zone rule is even introduced. Second, rank and percent rules disagree in 26.4% of modeled weeks under the same fan-support estimates, so rule choice is substantively important. Third, the judge-only baseline remains weak, matching only 36.4% of historical eliminated sets. The threshold mainly controls how much explicit judge discretion producers want in close calls; it does not change the underlying evidence that close-call transparency is needed.

12 Memo to DWTS Producers

To: DWTS Producers

Subject: A transparent voting system for close eliminations

The analysis suggests that fan votes should remain central to the show, but close bottom-two cases should not be treated as if the hidden fan vote were fully observable. We recommend replacing opaque rank aggregation with a Percent + Gray-Zone Judge Save system.

- Convert judge scores and fan votes into percentages before combining them. This makes the rule easier to explain to viewers.
- Use the standard percent result when the bottom-two margin is clearly separated.
- When the bottom-two margin is inside the gray zone, ask judges to save one contestant from the bottom two.

- Publicly describe the judge-save condition as a close-call safeguard, not as a replacement for fan influence.

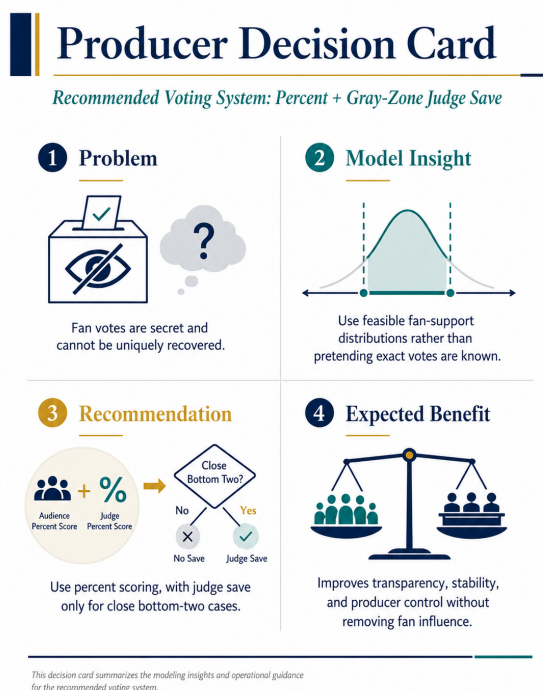


Figure 17: Producer decision card summarizing the recommended voting system.

13 Strengths, Limitations, and Extensions

Strengths. The model respects the fact that fan votes are hidden. It does not invent labels or claim to recover exact vote totals. Every numerical result in the paper is generated from the same reproducible workflow, and the validation step confirms that the reconstructed constraints are internally consistent with the observed modeled eliminations.

Limitations. The maximum-entropy fan profile is a conservative point estimate, not a direct observation. The bottom-two judge-save simulation is a policy model because detailed historical judge-save ballots are not fully available in the data. The professional-dancer and celebrity-effect regressions are observational associations and may reflect unmodeled casting, editing, or audience factors.

Extensions. Future work could add verified external popularity signals, such as social media engagement or search interest, but only if those sources are logged and validated. Another extension would model the evolution of fan support across weeks with a state-space model. We did not use those extensions in the main workflow because the official data already support a complete and transparent constraint-based solution.

References

- [1] COMAP. 2026 MCM Problem C: Data With the Stars. Mathematical Contest in Modeling, 2026.
- [2] COMAP. Contest Rules, Registration and Instructions for MCM/ICM. <https://contest.comap.com/undergraduate/contests/mcm/instructions.php>, 2026.
- [3] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.
- [4] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.
- [5] S. Diamond and S. Boyd. CVXPY: A Python-embedded modeling language for convex optimization. *Journal of Machine Learning Research*, 17(83):1–5, 2016.
- [6] P. Virtanen et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.
- [7] S. Seabold and J. Perktold. Statsmodels: Econometric and statistical modeling with Python. Proceedings of the 9th Python in Science Conference, 2010.